Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q19 (v1)

Two repeating patterns together: find the block where both start again, count in it, multiply

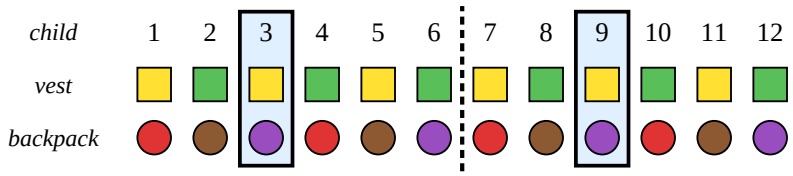
How to think about it: two patterns at the same time

In these questions two things repeat at the same time, for example the colour of the vest and the colour of the backpack. You have to find out how often a certain pair happens, like "yellow vest **and** purple backpack".

The big idea: two patterns together make one **longer pattern**. Find the block where **both start again at the same time**, count inside one block, then multiply.

Step by step: the contest question

- **1. Write it down.** Write the numbers 1, 2, 3, ... and draw both patterns under them. Stop when both patterns start again **at the same time**: child 7 looks exactly like child 1.

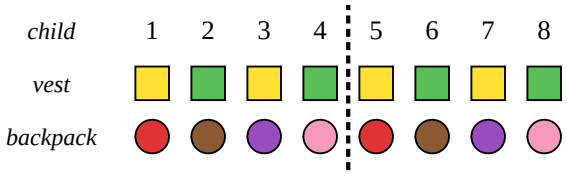


- **2. The block is 6 children long.** After every 6 children everything repeats (the dashed line).
- **3. Count in one block.** In children 1 to 6, only child 3 has a yellow vest and a purple backpack. So there is **1 in every block**.
- **4. Multiply.** 60 children = 60 ÷ 6 = 10 blocks. 10 × 1 = **10** children.

Another way to see it: yellow vests are children 1, 3, 5, 7, ... (odd numbers). Purple backpacks are children 3, 6, 9, 12, ... Both: 3, 9, 15, 21, ... always 6 more.

Watch out

- **The block is not always "one length × the other".** Vests with 2 colours and backpacks with 4 colours start again together after only **4** children, not 8. Then some pairs **never** happen. Here a green vest never gets a red backpack:

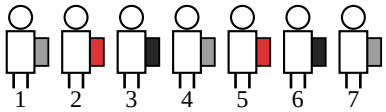


- **Children left over:** 64 children with a block of 6 is 10 blocks (60 children) and 4 more. Check those 4 one by one.

Try it yourself: take coloured pencils and draw the first 12 children of a pattern. Circle the ones you are looking for. Can you see the block?

Warm-up: small steps first (no points)

W1. Children stand in a row. Their backpacks repeat like this: grey, red, black, grey, red, black, ... What colour is the backpack of child number 12?

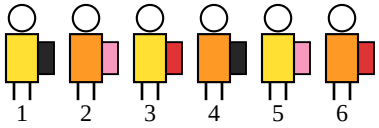


A. grey ***B.*** red ***C.*** black ***D.*** brown ***E.*** pink

W2. 24 children stand in a row. Their backpacks repeat like this: purple, grey, black, purple, grey, black, ... How many have a grey backpack?

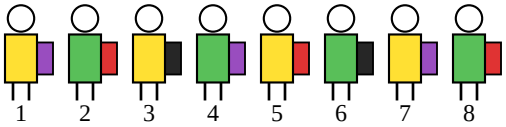
A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

W3. Here are the first 6 children of a row. How many of them have a **yellow** vest **and** a **red** backpack?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 6

W4. Look at the children. Child 1 has a yellow vest and a purple backpack. Which is the **next** child with a yellow vest and a purple backpack?



A. child 7 ***B.*** child 8 ***C.*** child 9 ***D.*** child 10 ***E.*** child 11

W5. In a row of 60 children, the children with a yellow vest and a purple backpack have the numbers 2, 8, 14, 20, ... (always 6 more). How many of them are there?

A. 6 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 20

Part A: Each question is worth 3 points

1. 60 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, green, yellow, green, ... The backpacks repeat like this: red, pink, black, red, pink, black, ... How many children have a **green** vest **and** a **black** backpack?

A. 5 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 12

2. 70 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, green, yellow, green, ... The backpacks repeat like this: red, grey, black, purple, pink, red, grey, ... How many children have a **green** vest **and** a **grey** backpack?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

Part B: Each question is worth 4 points

3. 50 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: blue, yellow, blue, yellow, ... The backpacks repeat like this: purple, black, pink, purple, black, pink, ... How many children have a **blue** vest **and** a **purple** backpack?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 16

4. 73 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: blue, green, orange, blue, green, orange, ... The backpacks repeat like this: pink, purple, red, black, brown, pink, purple, ... How many children have a **green** vest **and** a **brown** backpack?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 14 ***E.*** 24

5. 60 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, blue, yellow, blue, ... The backpacks repeat like this: red, pink, grey, red, pink, grey, ... What does child number **54** wear?

A. blue vest, grey backpack ***B.*** yellow vest, red backpack ***C.*** yellow vest, grey backpack
D. blue vest, pink backpack ***E.*** blue vest, red backpack

6. 100 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: orange, blue, yellow, orange, blue, yellow, ... The backpacks repeat like this: grey, purple, black, brown, grey, purple, ... What does child number **94** wear?

A. blue vest, purple backpack ***B.*** orange vest, purple backpack ***C.*** orange vest, black backpack
D. blue vest, black backpack ***E.*** yellow vest, purple backpack

7. 100 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: green, orange, green, orange, ... The backpacks repeat like this: pink, red, brown, pink, red, brown, ... Count only the children with a **green** vest and a **red** backpack. Which number in the row does the **5th** of them have?

A. 23 ***B.*** 25 ***C.*** 28 ***D.*** 29 ***E.*** 30

8. 100 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: green, orange, green, orange, ... The backpacks repeat like this: grey, black, pink, red, purple, grey, black, ... Count only the children with a **orange** vest and a **purple** backpack. Which number in the row does the **6th** of them have?

A. 60 ***B.*** 61 ***C.*** 65 ***D.*** 70 ***E.*** 72

Part C: Each question is worth 5 points

9. 28 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, green, yellow, green, ... The backpacks repeat like this: red, grey, purple, brown, red, grey, ... How many children have a **green** vest **and** a **red** backpack?

A. 0 ***B.*** 3 ***C.*** 6 ***D.*** 7 ***E.*** 14

10. 24 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: blue, yellow, orange, green, blue, yellow, ... The backpacks repeat like this: brown, purple, brown, purple, ... How many children have a **green** vest **and** a **brown** backpack?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 6 ***E.*** 12

11. A red lamp flashes every 2 seconds and a blue lamp flashes every 5 seconds. They flash together right at the start. How many times do they flash **together** in the next 50 seconds? (Count a flash exactly at second 50, but not the one at the start.)

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

12. A red lamp flashes every 4 seconds and a blue lamp flashes every 6 seconds. They flash together right at the start. How many times do they flash **together** in the next 72 seconds? (Count a flash exactly at second 72, but not the one at the start.)

A. 6 ***B.*** 7 ***C.*** 12 ***D.*** 18 ***E.*** 30

Answer Key

W1.	**C**	**W2.**	**E**	**W3.**	**B**	**W4.**	**A**	**W5.**	**D**	**1.**	**D**
2.	**E**	**3.**	**D**	**4.**	**A**	**5.**	**A**	**6.**	**B**	**7.**	**D**
8.	**A**	**9.**	**A**	**10.**	**A**	**11.**	**E**	**12.**	**A**		

Solutions

W1. C (black) — The colours repeat every 3 children. 12 = 3 × 3 + 3, so it is the same as child 3: black. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_one_pattern]

W2. E (8) — In every 3 children there is one grey backpack. 24 ÷ 3 = 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_how_many_one]

W3. B (1) — Look at each child: only child number 3. So 1. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_first_block]

W4. A (child 7) — The vest pattern starts again at 3, 5, 7, ... and the backpack pattern at 4, 7, 10, ... Both together: child 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_same_again]

W5. D (10) — One in every block of 6: 60 ÷ 6 = 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_skip_count]

1. D (10) — Both patterns start again together every 6 children. In each block of 6 only child number 6 (then 12, 18, ...) has a green vest and a black backpack. 60 ÷ 6 = 10 blocks, so 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

2. E (7) — Both patterns start again together every 10 children. In each block of 10 only child number 2 (then 12, 22, ...) has a green vest and a grey backpack. 70 ÷ 10 = 7 blocks, so 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

3. D (9) — The patterns start again together every 6 children; in each block child number 1 matches. 50 = 8 × 6 + 2: 8 full blocks give 8, and in the 2 children left over child 49 matches too. 8 + 1 = 9. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

4. A (5) — The patterns start again together every 15 children; in each block child number 5 matches. 73 = 4 × 15 + 13: 4 full blocks give 4, and in the 13 children left over child 65 matches too. 4 + 1 = 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

5. A (blue vest, grey backpack) — The vests repeat every 2: 54 = 26 × 2 + 2, so the vest is the 2nd colour: blue. The backpacks repeat every 3: 54 = 17 × 3 + 3, so the backpack is grey. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

6. B (orange vest, purple backpack) — The vests repeat every 3: 94 = 31 × 3 + 1, so the vest is the 1st colour: orange. The backpacks repeat every 4: 94 = 23 × 4 + 2, so the backpack is purple. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

7. D (29) — The first one is child 5. The patterns start again together every 6 children, so the next ones are 11, 17, ... The 5th: 5 + 4 × 6 = 29. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · kth_match]

8. A (60) — The first one is child 10. The patterns start again together every 10 children, so the next ones are 20, 30, ... The 6th: 10 + 5 × 10 = 60. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · kth_match]

9. A (0) — Write the first children down: both patterns start again together already after 4 children (not 2 × 4 = 8). In each block of 4 no child has a green vest and a red backpack, so the answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

10. A (0) — Write the first children down: both patterns start again together already after 4 children (not 4 × 2 = 8). In each block of 4 no child has a green vest and a brown backpack, so the answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

11. E (5) — Red flashes at 2, 4, 6, ... and blue at 5, 10, 15, ... The first time both flash is at second 10, then every 10 seconds. $50 \div 10 = 5$. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]

12. A (6) — Red flashes at 4, 8, 12, ... and blue at 6, 12, 18, ... The first time both flash is at second 12 (not $4 \times 6 = 24$), then every 12 seconds. $72 \div 12 = 6$. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]